

INSTRUCTION FORMATS

The Z80 instructions are listed in Chapter 4. Z80 instructions may be formatted in one, two, three or four bytes. An instruction specifies the operation to be performed by the microprocessor. From a simplified standpoint, every instruction may be represented as an opcode followed by an optional literal or address field, comprising one or two words. The opcode field specifies the operation to be carried out. In strict computer terminology, the opcode represents only those bits which specify the operation to be performed, exclusive of the register pointers that might be necessary. In the microprocessor world, it is convenient to call opcode the operation code itself, as well as any register pointers which it might incorporate. This "generalized opcode" must reside in an eight-bit word for efficiency (this is the limiting factor on the number of instructions available in a microprocessor).

The 8080 uses instructions which may be one, two, three, bytes long (see Figure 2.15). However, the Z80 is equipped with additional indexed instructions, which require one more byte. In the case of the Z80, opcodes are, in general, one byte long, except for special instructions which require a two-byte opcode.

Some instructions require that one byte of data follow the opcode. In such a case, the instruction will be a two-byte instruction, the second byte of which is data (except for indexing, which adds an extra byte).

In other cases, the instruction might require the specification of an address. An address requires 16 bits and, therefore, two bytes. In that case, the instruction will be a three-byte or a four-byte instruction.

For each byte of the instruction, the control unit will have to perform a memory fetch, which will require four clock cycles. The shorter the instruction, the faster the execution.

A One-Word Instruction

One-word instructions are, in principle, fastest and are favored by the programmer. A typical such instruction for the Z80 is:

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LD r, r'
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This instruction means: "Transfer the contents of register r' into r." This is a typical "register-to-register" operation. Every microprocessor must be equipped with such instructions, which allow the programmer to transfer information from any of the machine's registers into another one. Instructions referencing special registers of the machine,